

b) Let exactly one of the inscriptions be True.

Let Trunk 1 be True. $\therefore$ It is empty

Since Trunk 2 states There is a treasure in Trunk 1, it is False and consistent

Since Trunk 3 is False, There is no treasure in Trunk 2.

$\therefore$ The treasure is in Trunk 3

Let Trunk 2 be True. $\therefore$ There is a treasure in Trunk 1

Trunk 1 states that it is empty which is False and consistent.

Trunk 3 states that there is a treasure in Trunk 2 which is False and consistent

$\therefore$ The treasure is in Trunk 1

Let Trunk 3 be True. $\therefore$ There is a treasure in Trunk 2

Trunk 1 states that it is empty which is True. This is a contradiction to the fact that it must be False. $\therefore$ Trunk 3 is False

$\therefore$ The Queen can say this but the treasure is in either Trunk 1 or Trunk 3

(c) Let exactly two of the inscriptions are True.

Let Trunk 1 be False. $\therefore$ The treasure is in Trunk 1.

Trunk 2 states there is a treasure in Trunk 1 which is True and consistent.

Trunk 3 states that there is a treasure in Trunk 2 which is False. This is a contradiction to the fact that it must be True.

$\therefore$ Trunk 1 is True. The treasure is not in Trunk 1

Trunk 2 states that there is a treasure in Trunk 1 which must be False.

So Trunk 3 must be True. $\therefore$ The treasure is in Trunk 2

$\therefore$ The Queen can say this and the treasure is in Trunk 2

d) Let all three inscriptions be True.

Trunk 2 and Trunk 3 contradict each other by respectively stating that the treasure is in Trunk 1 and the Treasure is in Trunk 2.